Investigating the effect of two metabolic inhibitors on angiogenesis

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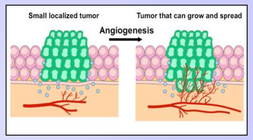
Introduction

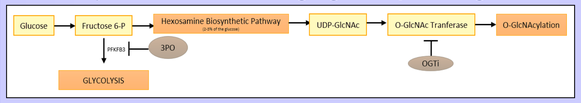
Endothelial cells (ECs) line the interior surface of blood vessels and drive angiogenesis, the process of new blood vessel formation from existing blood vessels. Angiogenesis is thought to be a critical factor in tumor growth [1].

Angiogenesis occurs through a complex, 3D process of endothelial proliferation, migration, and eventual maturation into a blood vessel.



Recently, EC metabolic activity has been shown to be important to angiogenic sprouting [2]. For this project, we decided to test two metabolic inhibitors, namely 3PO and OGTi, in an endothelial tube formation assay to determine their effects on angiogenesis. 3PO is known to inhibit glycolytic activity, and OGTi decreases protein O-GlcNAcylation through the hexosamine biosynthetic pathway.

The goal of this study was to evaluate the effect of these two inhibitors and understand the use of these inhibitory mechanisms in angiogenic sprouting in ECs.



Hypothesis

To determine the effect of 3PO and OGTi on an *in vitro* Endothelial pseudo vascular network model

Methods

HUVEC Tube Formation Assay

Pre-treatment

- Human umbilical vein endothelial cells (HUVECs) were treated with either 1 or 5 µM 3PO, or 1 or 10 µM OGTi for 24 hours. DMSO was the loading control.
- ~100,000 human umbilical vein endothelial cells (HUVEC) were seeded on 40 µl growth factor reduced Matrigel (Corning) in serum-free Endothelial Basal Media-2 (EBM-2, Lonza) supplemented with 10 ng/ml VEGF in an 8-well chamber cassette (Falcon).
- HUVEC network formation was imaged after 24 and 48 hours using a phase contrast microscope. Two images were acquired per treatment and network parameters, specifically junctions, segments, and branches, were quantified using the Angiogenesis Analyzer plugin in ImageJ software.

Post-Treatment

- The HUVEC network was formed as described.
- After 6 hours, HUVEC networks were treated with varying inhibitor concentrations.
- HUVEC network formation was imaged and analyzed after 24 and 48 hours as described

Results

A) Pre-treatment

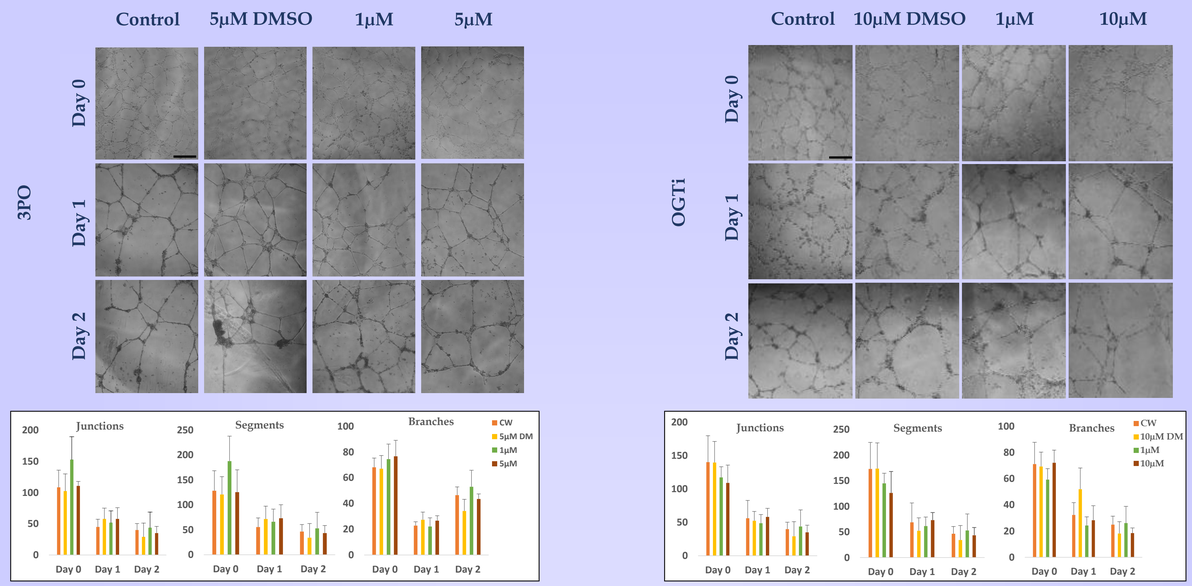


Figure 1. HUVEC tubes were formed as described. Here, the HUVECs were treated 24 hours before seeding. Sample size was 2 per condition. Samples were imaged by phase contrast microscopy on days 0, 1 and 2. Image J was used to analyze the number of junctions, branches and segments. Scale bar = 500µm.

B) Post-treatment

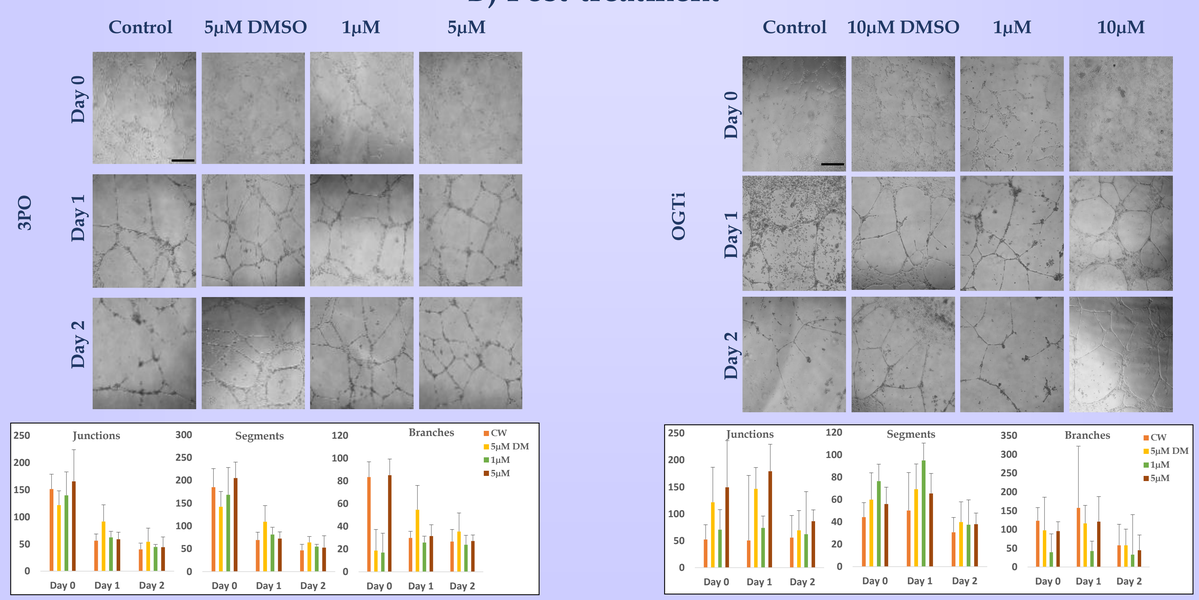


Figure 2. HUVEC tubes were formed as described. Here, the HUVECs were treated 24 hours before seeding. Sample size was 2 per condition. Samwere imaged by phase contrast microscopy on days 0, 1 and 2. Image J was used to analyze the number of junctions, branches and segments. Scale bar = 500µm.

Conclusions

- No significant difference in sprouting patterns was observed between the controls and the wells treated with the inhibitors.
- The results for both Pre-seeding and post-seeding treatment of HUVECs showed similar figures.
- The HUVEC tube formation assay may not be the most suitable assay to observe metabolic inhibition of angiogenesis.

Future Work

- To test inhibitory effects of 3PO and OGTi in increased concentrations.
- Investigate the effect of these metabolic inhibitors on angiogenesis in an aortic ring assay.
- Observe how angiogenic sprouting is inhibited when endothelial cells are seeded to form 3-D spheroids.
- Identify and test other potential angiogenic-targeted inhibitors in HUVECs assays.

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References

[1] Folkman, J. and Haudenschild, C., Angiogenesis in vitro. *Nature*, 1980, *288*(5791), p.551.

[2] Swaminathan, S. et al. Vascular Endothelial–Breast Epithelial Cell Coculture Model Created from 3D Cell Structures. *ACS Biomaterials Science & Engineering*, 2017, *3*(11), pp.2999-3006.